

Array:-

It is a user define datatype ~~this~~^{which} is used to store more than one value, within a single variable. It can store in a contiguous memory location that is within a array type variable more than one value can be stored in a sequence of memory location.

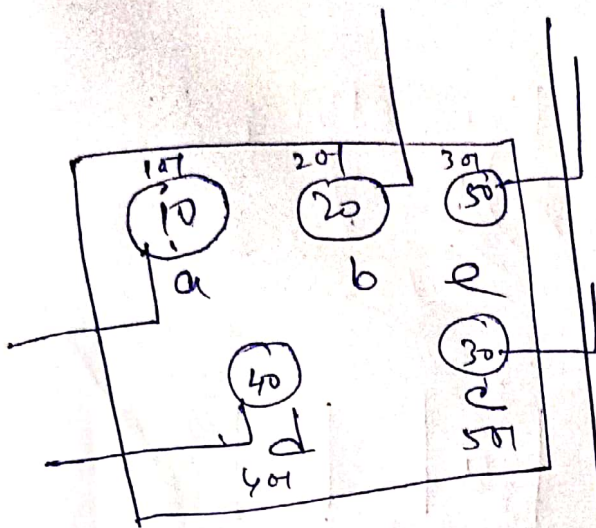
Integer type of array can only store integer type of value within it, like wise float type of array character type of array can store float and character value respectively. Array's ~~start~~ starting index will always be 0.

Variable⁺ within it

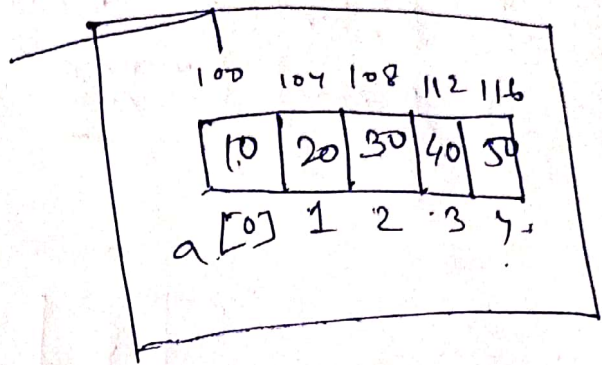
1. It can store a single value within it.
2. ~~Ex~~ Suppose we want to store 5 value. In that case minimum 5 variable is required.
3. As variables are stored in a haphazard memory location searching will take more times than array.
4. ~~Sorting~~ Sorting can't be possible because data is not sorted in a ascending or descending order format.
5. As minimum 5 variables ~~is~~^{are} required so minimum 5 reference is required which is time consuming to search and sort.

Array

1. It can store more than one value within it.
2. In case of Array only a single variable of array type can store five value within it.
3. As array's variable are stored in a contiguous memory location so, to search any element will take very less time than variable.
4. Sorting can't be possible as ~~variables~~^{array's} variables stored in a contiguous memory location.
5. As maximum 1 variable is required to store 5 value so only 1 reference is needed to ~~store~~ search and sort.



variable
(RAM)



Array
(RAM)

Syntax :-

`int a[5];`

`int a[n];`
`for(i=0; i < 5; i++)`

Static
Array



`for(i=0; i < 2; i++)`

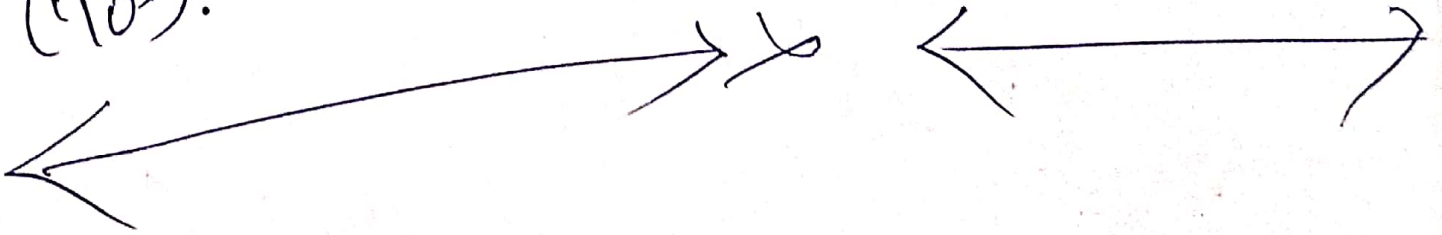
`scanf("%d", &a[i]);`

```
char X[5];  
printf("Enter the string");  
scanf("%s", &X);
```

0	1	2		
p	i	a	x	x

piu
an
Piya

within character array last character will ~~be~~ null
('\0').



char x[5];

printf("Enter the string = ");

scanf("%s", &x);

for (i=0; x[i] != '\0'; i++)

{ printf("%c", x[i]); }

RAM

A	v	i	\0
0	1	2	3

i = 3

Avi

x[i]

value of ith index at x array

J	o	y	i	t	a	\0
0	1	2	3	4	5	6

for (i=0; i<5; i++)

Avi

x[3] = '\0'

```

for (i = i - 1; i >= 0; i--)
{
    printf("%c", x[i]);
}

```

②

1	2	3	4	5	6	7
s	n	e	h	a	.	a

①

s	a	k	l
8	9	10	11

i = 10

kasab-ahens

s	n	e	h	a	\0
---	---	---	---	---	----

x

	0	1	2	3	4	5
y	s	n	e	h	a	\0

5
4
3
2
1
0

for (i=0; ~~X[i] != '\0'; i++~~)

{

printf("%c", X[i]);

Y[i] = X[i];

}

Y[i] = '\0';

sneha